

Research V8R Recovery Evidence for PID and Fuzzy PID Manipulator Control

Research-v8R recovery evidence

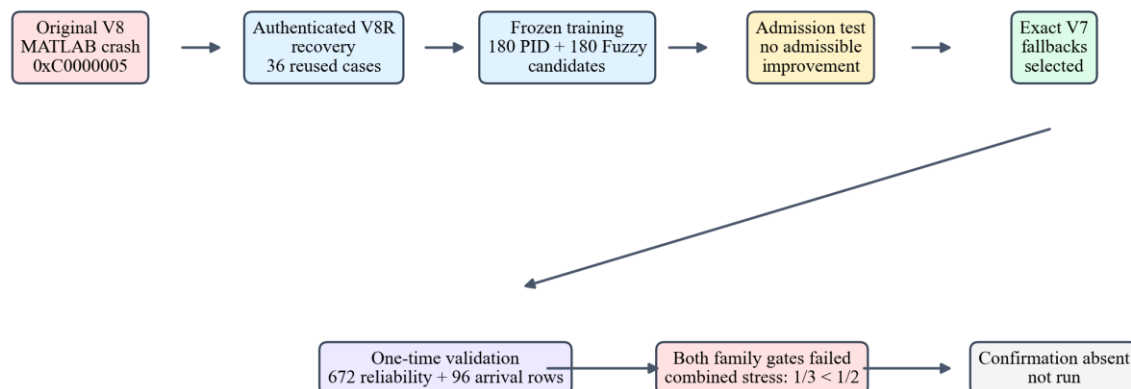
Abstract and research question

The recovery study did not find a newly admissible controller improvement. After independent searches, both families selected exact V7 fallbacks. One-time validation recorded PID 55/112 and Fuzzy-PID 57/112 requested-design reliable successes. Each family achieved 1/3 in the feasible-executed combined-stress cell, below the required 1/2; confirmation was not run.

We ask whether parameter tuning can improve reliability and stable arrival under fixed trajectory and command-generation rules without increasing failures or exceeding the permitted actuation cost. The study retains a given-path task and an endpoint-only task. It compares position histories, stable arrival and applied mechanical work, rather than optimizing planning speed in the core experiment.

The contribution is an authenticated recovery procedure and a negative admission result with inspectable simulation histories. These findings support a new preregistered experiment, not a guarantee of high success or a hardware claim.

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All arrows describe authenticated evidence flow; no simulation was rerun for this figure.

Figure 1. Evidence flow from the interrupted V8 attempt to frozen V8R validation. The confirmation branch stopped at the failed family gate.

System model and frozen controllers

The planar two-link nonlinear plant includes configuration-dependent inertia, Coriolis and centrifugal torque, gravity, viscous friction, payload and external disturbance. Its kinematics and joint-space control follow the conventional manipulator formulation [1]. Feasibility checks use the prescribed reference and the same scenario torque limits as execution.

$$M(q) \ddot{q} + C(q, \dot{q}) \dot{q} + G(q) + B \dot{q} = \tau + \tau_{dist}$$

Manual PID uses independent joint feedback. Original Fuzzy-PID uses a Mamdani inference system to update proportional, integral and derivative gains from scaled error and error rate [2,3]. The selected reliability controllers inherit the frozen V7 model compensation, derivative filtering and anti-windup implementation; their V8R hashes are exact duplicates of their corresponding V7 contracts. Base gains below are rounded to three significant digits, not instantaneous fuzzy gains.

Controller	Kp joints 1 and 2	Ki joints 1 and 2	Kd joints 1 and 2
Manual PID	120, 100	40, 30	25, 18
Original Fuzzy PID	120, 100	40, 30	25, 18
V8R PID fallback	62.8, 117	39.3, 17.8	14, 15
V8R Fuzzy fallback	120, 100	40, 30	25, 18

All core tasks keep trajectory or command generation fixed. The path mode is a Cartesian quintic command converted to a continuous joint reference. The point mode applies the final joint target immediately from the same initial state. The endpoint-only transient is not constrained to follow the Cartesian line.

Interrupted V8 and authenticated recovery

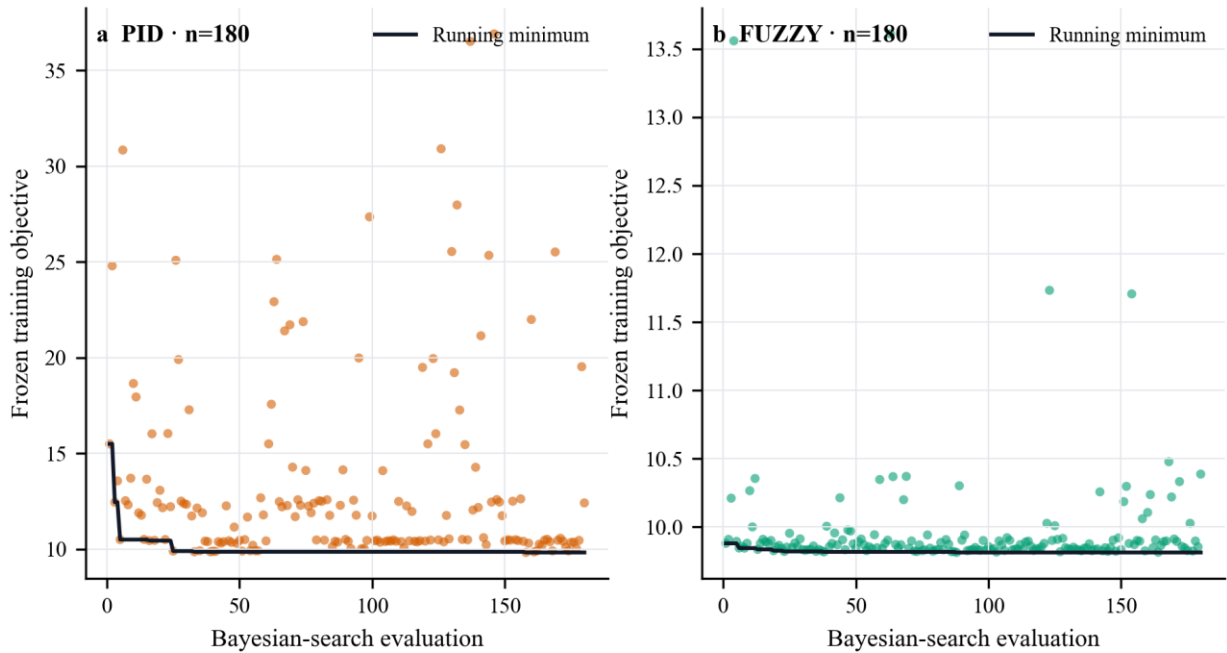
The sealed recovery manifest records the original V8 access violation 0xC0000005 before a training seal and authentication and recomputation of 36 checkpoint candidates with zero non-identity metric mismatches before reuse. The original attempt marker and checkpoint remain unchanged. This communication layer independently verifies unchanged sealed input bytes, not the historical crash or candidate recomputation. Post-recovery search removed redundant controller execution and unnecessary history capture; final evaluations still retained all histories. This resource reduction does not establish the native crash cause or prove future crashes impossible.

Training search and selection

The searches retained 180 PID evaluations and 180 Fuzzy evaluations in total across three fixed seeds per family. Bayesian optimization proposes expensive simulation evaluations [4]; proposal score and final pool admission are separate decisions. The same frozen reliability and paired arrival design is used for every candidate within a family.

The V8 objective inherits the V7 seven-component case score. Nonfinite components are assigned 10 and negative components are clipped to zero. The weights for joint RMS, steady maximum, EE maximum, recovery failure, saturation duty, slew and squared torque are respectively 0.25, 0.15, 0.20, 0.15, 0.10, 0.075 and 0.075. Case scores are averaged within reference clusters; the continuous aggregate weights mean, P90, CVaR95 and worst cluster score by 0.20, 0.30, 0.40 and 0.10. Feasible accepted failure and non-recovery fractions add penalties of 100 and 25. This is not the older research-v4 objective.

Pool admission requires no worsening in specified failures, limits, saturation and P90 errors, and no reduction in nominal, long-travel or either arrival-mode success count. Slew and squared-torque allowances are 110% of baseline. Admissible candidates rank lexicographically by combined-stress failures, capped arrival time and squared torque. Mechanical work is recorded, not optimized or used for selection.



Neither family produced an admissible improvement; both selected exact V7 fallbacks.

Figure 2. All retained candidate objectives by evaluation index. A lower proposal score alone does not imply a candidate passed admission.

Both manifest decisions are accepted=false with reason no-admissible-improvement. Selecting an exact V7 fallback is not a new optimization win.

Validation denominators and reliability

Requested counts every designed task (112 per controller), including planner rejection and unsuccessful execution. Feasible design requires ReferenceFeasible and ActuationFeasible (63). Executed records controller execution (92); it is not identical to physical feasibility. Feasible-executed requires all three fields (59).

ProtocolSuccess requires completed simulation, per-joint steady RMS at most 0.02 rad and steady maximum at most 0.05 rad in the final 0.5 s, and full-horizon EE maximum at most 0.15 m. ReliableSuccess additionally requires reference feasibility, actuation feasibility, no joint-limit violation and recovery pass. No-disturbance cases are not counted as non-recovery.

Controller	Reliable requests	Rate	Feasible executed	Wilson 95% request interval
Manual PID	14/112	12.5%	14/59	7.6 to 19.9%
Original Fuzzy PID	20/112	17.9%	20/59	11.9 to 26.0%
V8R PID fallback	55/112	49.1%	55/59	40.0 to 58.2%
V8R Fuzzy fallback	57/112	50.9%	57/59	41.8 to 60.0%

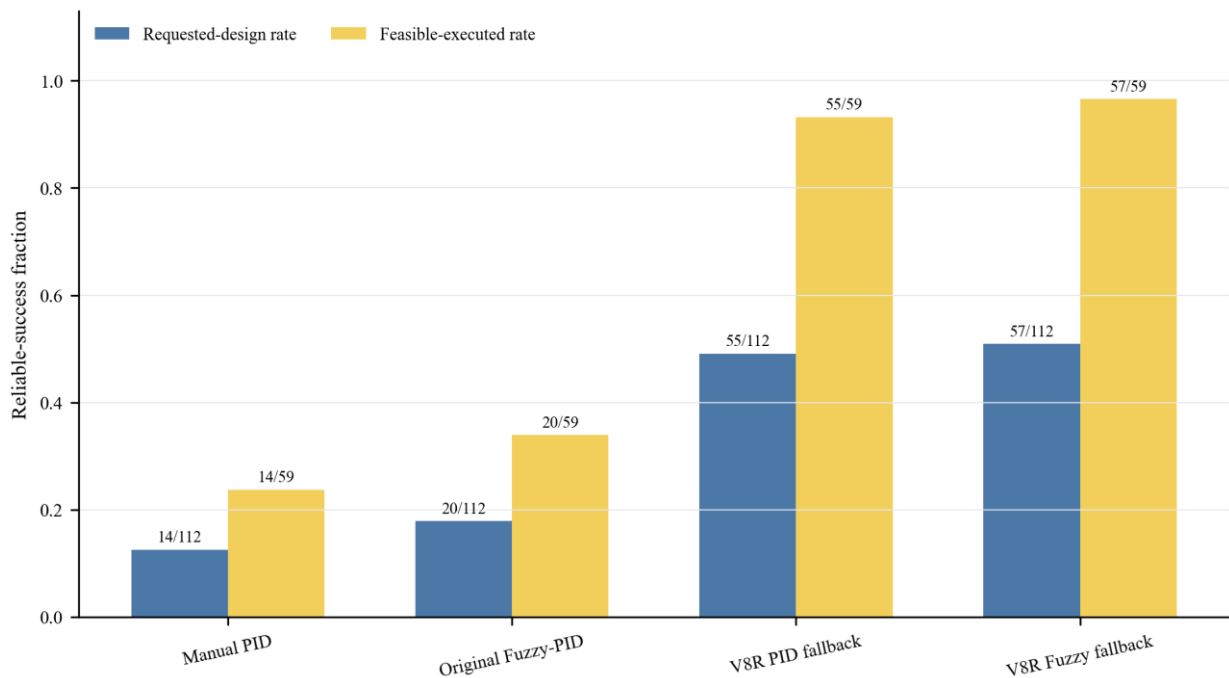


Figure 3. Requested-design and feasible-executed rates. Wilson intervals are reported in the table; the bars show point estimates.

Wilson intervals [5] are descriptive binomial summaries, not cluster-aware superiority tests. Shared trajectories and paired uncertainties are not asserted to be independent trials.

Condition reliability and feasibility

Condition summaries retain the frozen morphology, payload, inertia, noise, disturbance, torque and combined-stress assignments. The heatmap displays four unique controller contracts. Raw CSVs retain six labels, including both V7 references, whose outcomes equal their corresponding V8R fallbacks. Duplicated contracts are not treated as independent controllers.

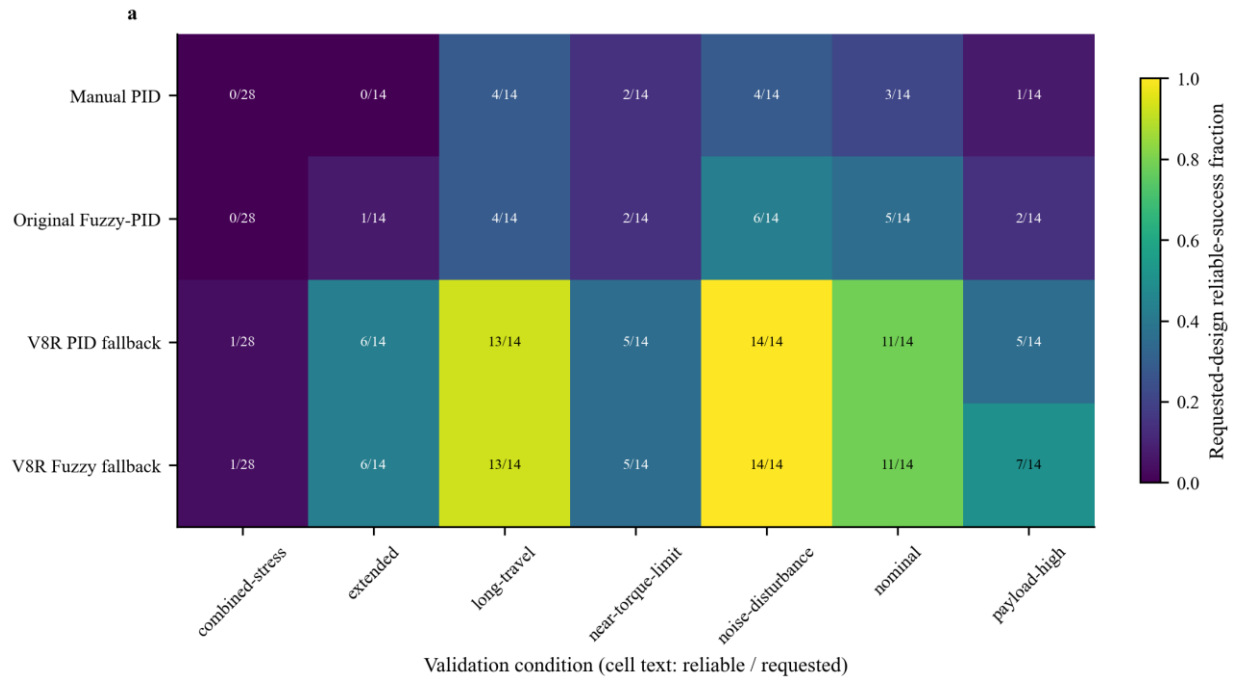


Figure 4. Requested-design reliable successes by condition. Cell annotations retain the numerator and requested denominator.

The most consequential gate cell is combined stress: both selected families record 1/3 in the feasible-executed subset. Its denominator is small. A condition-wise point estimate cannot establish a precise population probability, and the plot does not isolate a single causal mechanism.

A controller-conditional estimate asks how well an implementation executes physically feasible accepted inputs. The requested-design estimate instead includes the planner and physical envelope. Both are needed to avoid disguising an infeasible command as a successful control task.

Feasibility and overlapping failure labels

Reference feasibility checks finite prescribed motion and joint bounds. Actuation feasibility compares inverse-dynamics torque demand with the scenario torque limit; a demand ratio equal to one passes and a ratio above one fails. Slow retiming cannot eliminate static gravity demand that already exceeds available torque.

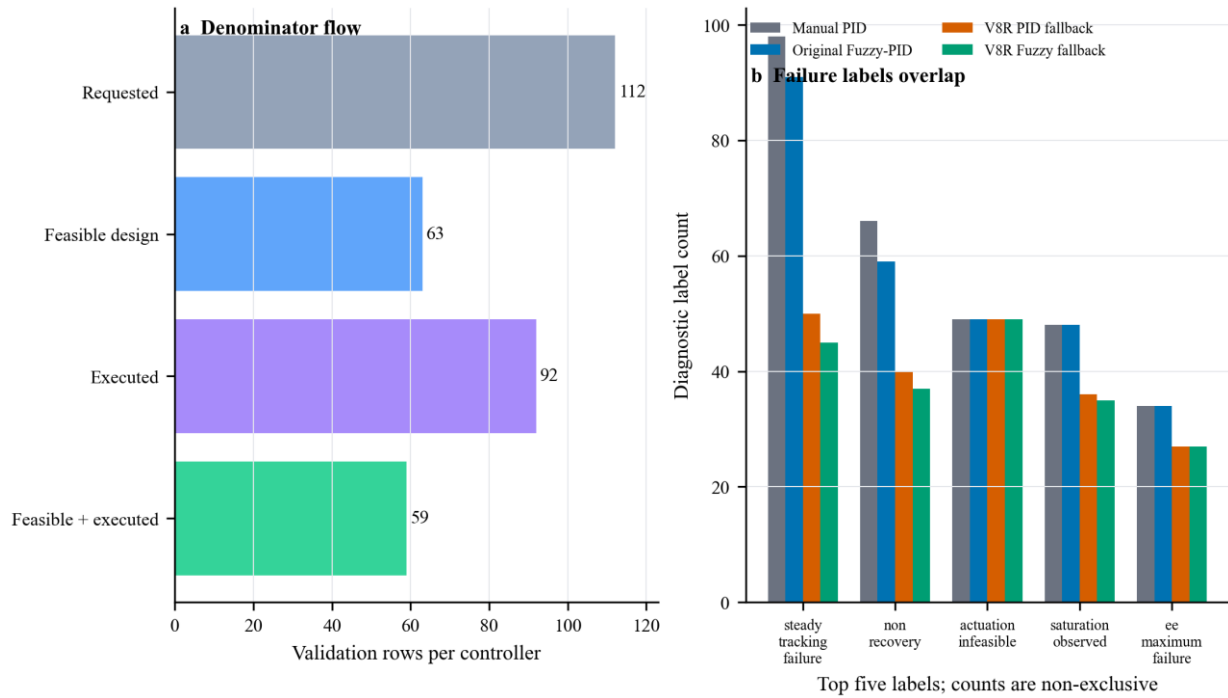


Figure 5. Physical feasibility and execution denominators alongside non-exclusive failure-label counts. One unsuccessful row may contribute to several labels.

The failure labels identify reference or actuation infeasibility, incomplete simulation, joint limits, steady tracking, EE maximum, non-recovery, saturation and planner disposition. Their counts must not be summed as mutually exclusive causes. A static-infeasible planner rejection is retained as a requested failure, without fabricating an executed robot history.

Saturation observed is a diagnostic flag, not sufficient on its own to prove a failure mechanism. Conversely, an accurate-looking motion trace does not override a failed full-horizon criterion. The distinction matters when judging visually similar controllers.

Tracking and disturbance recovery

The tracking example is v8-validation-0039. Selection maximizes total prescribed joint travel among design-feasible validation inputs, with lexical ID tie-breaking; no outcome is used. The panels show each joint's prescribed and actual position on the original time base. Display downsampling preserves boundary and recovery samples.

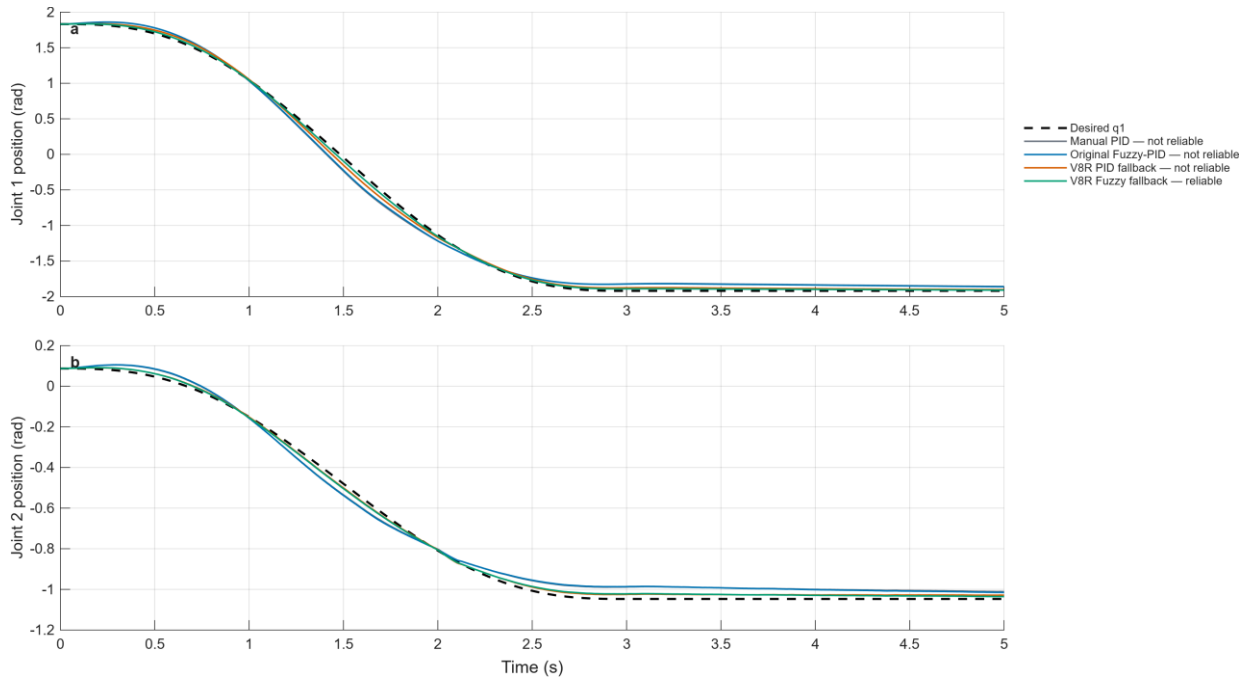


Figure 6. Desired and actual joint positions for an outcome-blind validation input. The legend reports the full-run reliable outcome, not an instantaneous visual judgment.

The curves are explanatory examples, not evidence that the displayed controller always wins. An EE maximum failure, recovery failure or joint limit can remain relevant even when final joint error is small. Aggregate rates remain controlled by every requested row.

The reliability video additionally uses the largest-travel feasible nominal input. Its fixed camera compares actual links against dashed prescribed links and retains controller differences without rescaling their dynamics.

Disturbance recovery and rejected stress input

The recovery plot uses the same executed validation input v8-validation-0039; it is not presented as the rejected combined-stress example. It overlays the disturbance window and the recovery band against the recorded joint error. Recovery requires a continuous dwell interval inside the joint-error band after disturbance ends, not merely a late low-error sample.

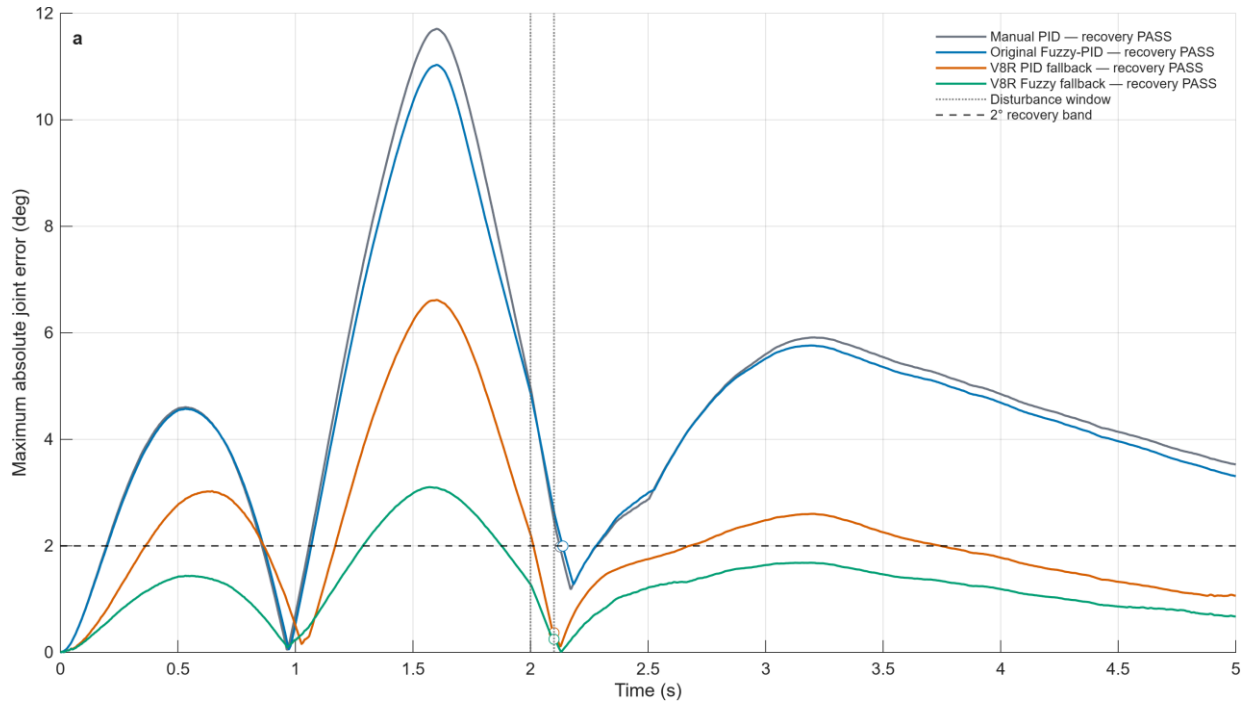


Figure 7. Recorded disturbance recovery in the selected executed case. Recovery pass is distinct from the complete ReliableSuccess conjunction.

The lexically selected combined-stress video chapter was planner-rejected-static-infeasible for all shown contracts. It therefore animates only the prescribed dashed input and labels NOT EXECUTED; there is no actual-motion claim. The rejection is evidence about the requested system envelope, not a measured recovery transient.

These views separate three outcomes that can otherwise be conflated: input rejection, failure to recover after executed disturbance, and failure of another tracking or safety criterion. No single-cause claim is inferred from their co-occurrence.

Given path and endpoint only arrival

The matched position example is v8-arrival-validation-draw-00247, chosen by largest prescribed start-to-target displacement and lexical tie-breaking. Both modes share endpoints, initial state, controller contracts and execution conditions. Command-generation rules are fixed; path speed and movement duration are not tuned in this core comparison.

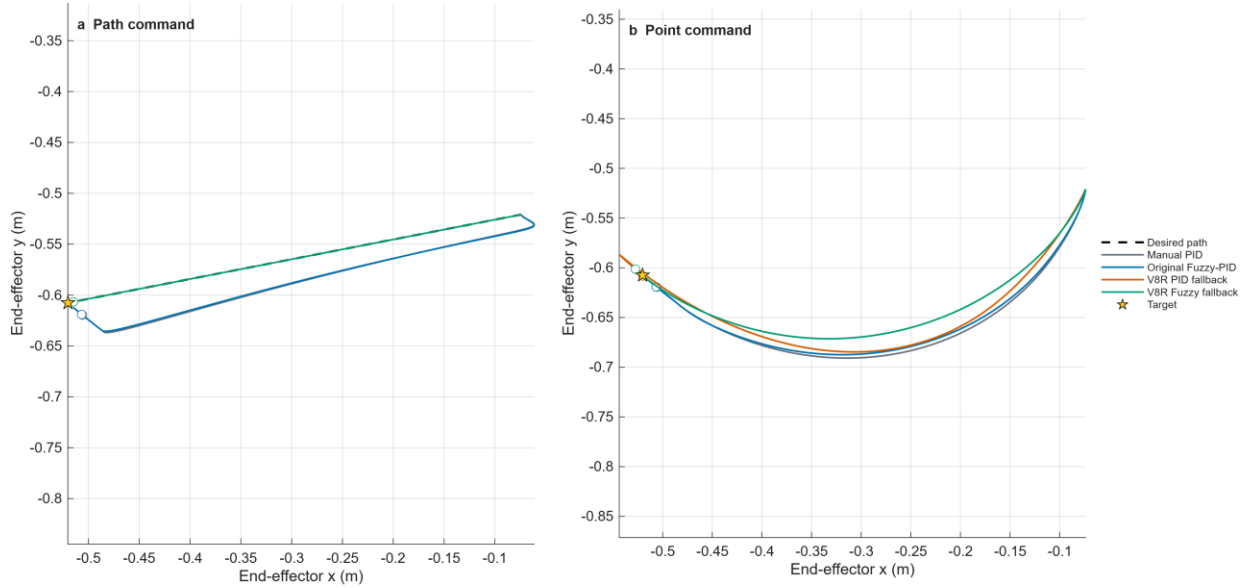


Figure 8. Actual XY trajectories for given-path and endpoint-only commands. Dashed commands and target markers distinguish prescribed motion from unconstrained transient motion.

StableArrivalTime is measured from command onset and records the end of the first continuous 0.20 s hold with end-effector distance at most 0.02 m. There is no velocity threshold in this frozen arrival criterion. SafeArrival additionally requires a complete 6 s horizon, feasible reference and actuation, finite state and actuation, no joint-limit violation and no observed late relapse outside the position band. A fast crossing without stable dwell is not promoted to a successful arrival.

Endpoint-only control may reach a stable target sooner because it is not obliged to track the timed Cartesian path. This compares two task definitions under frozen rules, not only two feedback algorithms. Mode-specific failures therefore remain visible.

Arrival distributions retain unsuccessful requests

Selected family	Command mode	Safe arrival	Safe only median time s
V8R PID fallback	path	8/8	2.5705
V8R PID fallback	point	7/8	0.5990
V8R Fuzzy fallback	path	8/8	2.5705
V8R Fuzzy fallback	point	6/8	0.6840

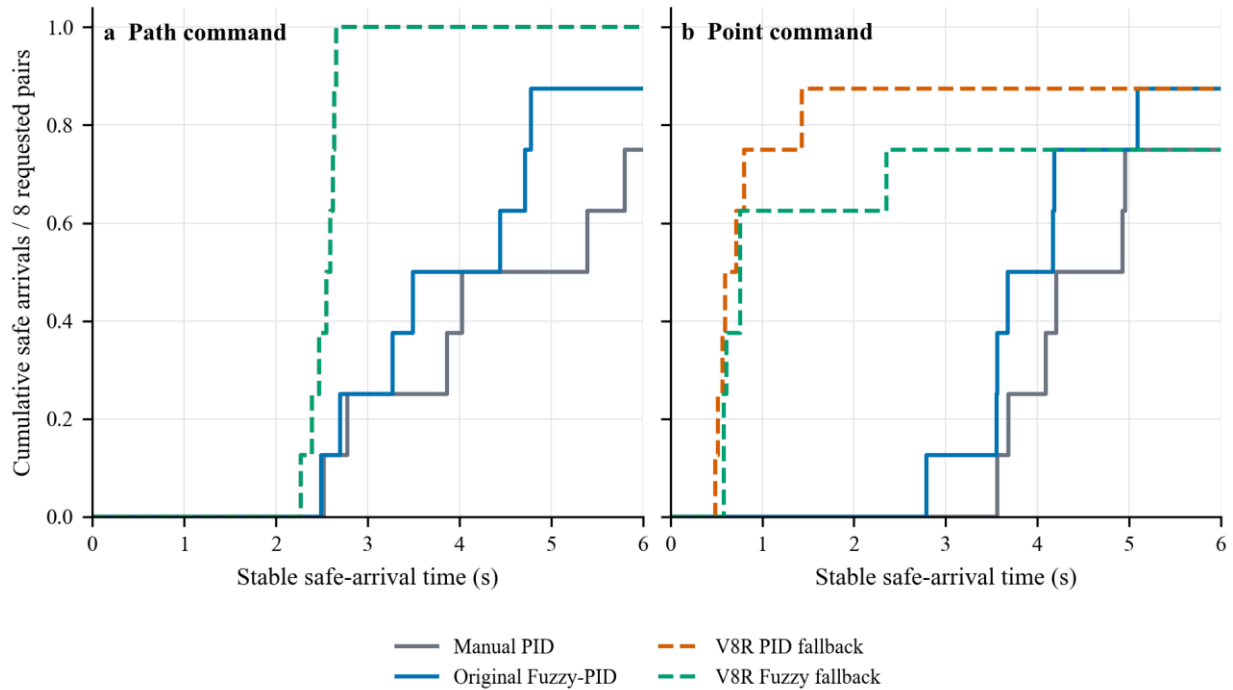


Figure 9. Cumulative safe completion divided by all requested pairs in each mode. Failed and non-arriving tasks remain in the denominator.

Median times in the table are conditional on safe arrival. They are not unconditional speed estimates: excluding failures can make a controller appear faster. The cumulative completion curves preserve this limitation by keeping all requests in the denominator.

The selected contracts are exact V7 fallbacks; each paired V8R-minus-V7 metric remains zero under the frozen validation protocol. A superiority test of identical contracts would not establish new tuning effectiveness. The two video chapters replay one matched pair only, while this figure summarizes every validation pair.

Mechanical work and torque cost

The mechanical-work metric is positive applied shaft work: integrate the sum of each joint's positive applied torque times joint velocity. Units are joules, with no regenerative credit. It is not electrical energy; motor efficiency, drive loss and battery consumption are not modelled.

Controller	Finite executed work cases	Median horizon work J
Manual PID	91	8.0413
Original Fuzzy PID	91	7.8794
V8R PID fallback	92	7.8841
V8R Fuzzy fallback	92	7.6115

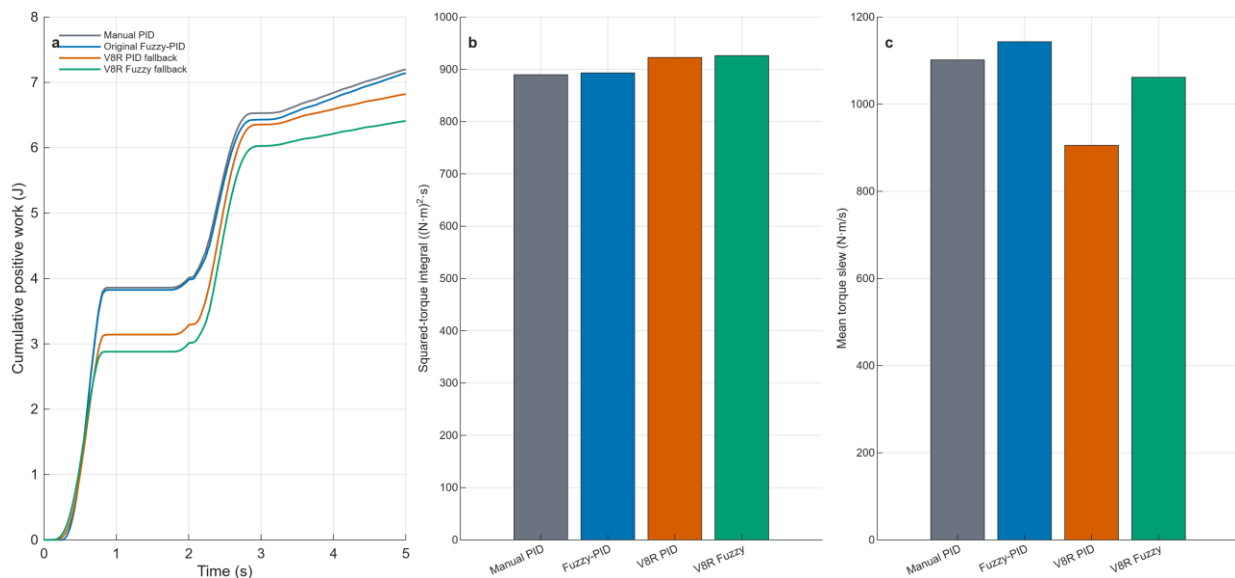


Figure 10. Cumulative positive mechanical work and applied-torque cost for the selected tracking input. These traces do not replace all-scenario summaries.

SquaredTorqueIntegral integrates the sum of squared applied joint torques, in $(N \cdot m)^2 \cdot s$. MeanTorqueSlew is the average of the joints' RMS torque derivatives, in $N \cdot m/s$. These are distinct actuation-cost measures and are not renamed energy. Finite work denominators are explicit; unsuccessful cases are not deleted from success denominators.

Training recorded mechanical work for later comparison but did not use it in objective or candidate ranking. A modest work difference cannot establish a tuning win when the selected controller is unchanged.

Gate decision and research limitations

Both family gates failed: the feasible-executed combined-stress rate $1/3$ is below $1/2$. Overall conditional execution and effort matched the exact V7 contracts, but that cannot override a failed condition floor. The manifest records confirmationAdmitted=false; confirmation was not run.

V8R improves evidence completeness and recovery traceability, not selected-controller performance. Noise amplification, conservative fuzzy rules and search dimensionality remain hypotheses unless separately tested. Neither controller is guaranteed to win, and simulation does not establish hardware reliability.

Recommendation for future research, not an implemented or empirically verified V9 protocol: a new training and held-out design, fixed core command rules and prespecified reliability targets. A future study must not recycle V8R validation as tuning feedback. Planning-speed sweeps belong to a separately labelled extension.

Reproducibility. The row-level CSVs, complete MAT histories, controller contracts and phase manifests are sealed under results/research-v8r. The new visual manifest binds raw inputs and presentation recipes; report and video hashes bind this communication layer. The archived trace recipe authenticates the original generation source; the current publication entry point is hardened. Byte-for-byte figure/video reproduction has not been performed. Every live video is labelled MATLAB nonlinear simulation - not physical footage.

Reliability simulation video

Arrival simulation video

References

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